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Federação Dinâmica de Redes e Fatiamento na Borda

Dynamic Network Federation and Edge Slicing

PROPOSTA DE TESE

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Dynamic Network Federation and Edge Slicing

Proposta de Tese apresentada à Universidade de Aveiro para cumprimento dos requisitos necessários à conclusão da unidade curricular Proposta de Tese, condição necessária para obtenção do grau de Doutor em Engenharia Informática, realizada sob a orientação científica do Doutor Daniel Corujo, Professor associado c/ agregação do Departamento de Eletrónica, Telecomunicações e Informática da Universidade de Aveiro, do Doutor Rui Aguiar, Professor catedrático do Departamento de Eletrónica, Telecomunicações e Informática da Universidade de Aveiro.

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Um sincero obrigado a todos que me aturam e inspiram, quer em regime presencial, remoto ou híbrido.

palavras-chave

6G, Federação de Rede, Computação de Borda Multi-Acesso, Fatias de Rede, Redes Privadas/Públicas

resumo

As redes 6G do futuro terão de integrar de forma transparente ambientes altamente heterogéneos, combinando infraestruturas de nuvem centralizadas com computação de borda extrema. Esta proposta de tese aborda os desafios da federação de redes multidomínio, focando-se na orquestração dinâmica de fatias de rede e na computação de borda multi-acesso (MEC) em domínios públicos e privados. Ao explorar cenários altamente exigentes - especificamente, estádios inteligentes de alta densidade e operações entre fronteiras de Proteção Pública e Socorro em Catástrofes (PPDR) - esta tese de doutoramento visa desenvolver mecanismos de federação seguros, escaláveis e sem estado. O trabalho irá propor uma arquitetura inovadora para a orquestração entre domínios, tirando partido das interfaces da Ligação Este-Oeste (EWBI) e de interligação programável para garantir a prestação contínua de serviços durante cargas de rede imprevisíveis e respostas a emergências críticas.

keywords

6G, Network Federation, Multi-Access Edge Computing (MEC), Network Slicing, Private/Public Networks

abstract

Future 6G networks will be required to seamlessly integrate highly heterogeneous environments, merging centralized cloud infrastructures with extreme edge computing. This thesis proposal addresses the challenges of multi-domain network federation, focusing on the dynamic orchestration of network slices and Multi-Access Edge Computing (MEC) across public and private domains. By exploring highly demanding scenarios - specifically high-density smart stadiums and cross-border Public Protection and Disaster Relief (PPDR) operations - this PhD thesis aims to develop secure, scalable, and stateless federation mechanisms. The work will propose a novel architecture for cross-domain orchestration, leveraging East-West Bound Interfaces (EWBI) and programmable interconnection to ensure continuous service delivery during unpredictable network loads and critical emergency responses.

acknowledgement of use of AI tools

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objetivos de desenvolvimento sustentável / **sustainable development goals**

Este trabalho contribui para os seguintes Objetivos de Desenvolvimento Sustentável (ODS) da Agenda 2030 das Nações Unidas (`https://ods.pt/`):

ODS 9 – Indústria, Inovação e Infraestruturas
ODS 11 – Cidades e Comunidades Sustentáveis

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Acronyms

3GPP	3rd Generation Partnership Project
5G	5th Generation
6G	6th Generation
AF	Application Function
AI	Artificial Intelligence
AI^2	Artificial Intelligence Integration
AMF	Access and Mobility Management Function
API	Application Programming Interface
AUSF	Authentication Server Function
BSS	Business Support System
CAMARA	Connected APIs for Mobile and Networking
CFS	Customer Facing System
CI/CD	Continuous Integration/Continuous Deployment
CNF	Cloud-native Network Function
COTS	Commercial Off-The-Shelf
CSMF	Communication Service Management Function
DNS	Domain Name System
eMBB	Enhanced Mobile Broadband
EMS	Element Management System
ETSI	European Telecommunications Standards Institute
EWBI	East-West Bound Interface
gNB	Next Generation Node B
GR	Group Report
GS	Group Specification
GSMA	Global System for Mobile Communications Association
IIoT	Industrial Internet of Things
IoT	Internet of Things
ISG	Industry Specification Group
KPI	Key Performance Indicator
LCM	Life Cycle Management
LLM	Large Language Model
MANO	Management and Orchestration
MCP	Model Context Protocol
MEAO	MEC Application Orchestrator
MEC	Multi-Access Edge Computing
MECF	MEC Federation
MEF	MEC Federator
MEH	MEC Host
MEO	MEC Orchestrator
MEP	MEC Platform
MEPM	MEC Platform Manager
ML	Machine Learning
mMTC	massive Machine-Type Communications
NBI	North Bound Interface
NEF	Network Exposure Function
NFV	Network Functions Virtualization
NFVI	NFV Infrastructure
NFVO	NFV Orchestrator
NRF	Network Repository Function
NSI	Network Slice Instance
NSMF	Network Slice Management Function
NSSAAF	Network Slice-Specific Authentication and Authorization Function
NSSAI	Network Slice Selection Assistance Information
NSSF	Network Slice Selection Function
NSSI	Network Slice Subnet Instance
NSSMF	Network Slice Subnet Management Function
NST	Network Slice Template
NTN	Non-Terrestrial Network
OP	Operator Platform
OPG	Operator Platform Group
OSS	Operation Support System
PCF	Policy Control Function
PoC	Proof of Concept
PPDR	Public Protection and Disaster Relief
QoS	Quality of Service
RAN	Radio Access Network
RIS	Reconfigurable Intelligent Surface
SBI	South Bound Interface
SD	Slice Differentiator
SDG	Software Development Group
SDN	Software Defined Networking
SLA	Service Level Agreement
SMF	Session Management Function
S-NSSAI	Single Network Slice Selection Assistance Information
SotA	State of the Art
SST	Slice/Service Type
UDM	Unified Data Management
UE	User Equipment
UNI	User Network Interface
UPF	User Plane Function

URLLC Ultra-Reliable Low-Latency Communications
V2X Vehicle-to-Everything
VI Virtual Infrastructure
VIM Virtual Infrastructure Manager
VNF Virtual Network Function
VNFM Virtual Network Functions Manager
WP Work Package
XR Extended Reality

CHAPTER 1

Introduction

The rapid evolution towards 5G-Advanced and beyond, culminating in the emerging vision of the 6th Generation (6G) of mobile networks, is fundamentally reshaping the telecommunications landscape. Future networks must serve not only traditional mobile broadband users, but an ever-expanding array of latency-sensitive, reliability-critical, and computationally intensive applications, including immersive Extended Reality (XR), autonomous vehicles, industrial automation and mission-critical public safety communications. Delivering this diversity of services requires a fundamental departure from the centralized, single-operator architecture that has historically defined cellular networks. Instead, future connectivity must emerge from a distributed, multi-provider continuum, where infrastructure, compute, and service capabilities are dynamically shared and coordinated across heterogeneous administrative domains. This shift introduces complex challenges around resource allocation, policy enforcement, and service continuity, which no single operator can address in isolation, and for which the industry still lacks mature, standardized solutions.

Technologies such as Network Functions Virtualization (NFV), Software Defined Networking (SDN), network slicing, and Multi-Access Edge Computing (MEC) have individually matured to support flexible, programmable network deployments. However, these technologies have predominantly been designed and deployed within the boundaries of a single administrative domain, offering insufficient support for the dynamic resource negotiation, cross-domain service chaining, and slice mobility that multi-operator services require. Emerging standards provide a promising foundation, but their integration into automated, end-to-end orchestration pipelines remains largely unexplored.

This thesis proposal outlines a research agenda centered on dynamic multi-domain network federation and edge slicing, addressing the gap between isolated single-domain deployments and true cross-operator service continuity. The proposed research will develop a stateless MEC slice mobility framework, treating slice components as portable microservice bundles decoupled from their origin domain, with inter-domain orchestration grounded in the East-West Bound Interface (EWBI) interface of the Global System for Mobile Communications Association (GSMA) Operator Platform Group (OPG) and augmented using intent-based management methods. The approach is planned to be evaluated against two scenarios: a smart stadium environment requiring elastic federation with public networks during crowd spikes, and a cross-border Public Protection and Disaster Relief (PPDR) deployment demanding strict Service Level Agreement (SLA) guarantees and operational continuity for mobile first-responders.

1.1 Document Structure

The remainder of this document is organized as follows. Chapter 2 surveys the key enabling technologies and standardization efforts underpinning this work, covering 5G and 6G network architectures, NFV, SDN, network slicing, MEC, intent-based orchestration, and multi-domain federation frameworks including the GSMA OPG model and the European Telecommunications Standards Institute (ETSI) MEC federation specifications. Chapter 3 defines the research questions, objectives, and expected contributions of the thesis, articulating the specific gaps addressed and the scientific novelty of the proposed approach. Chapter 4 details the workplan and schedule, describing the work packages, tasks, milestones, and deliverables structured across the thesis timeline.

CHAPTER 2

Background and Key Enablers

This chapter introduces the key technologies underpinning this work. It begins with 5G networks and their architectural evolution toward 6G, then covers NFV and SDN as the programmability facilitators, Network Slicing as the multi-service isolation mechanism, and MEC as the edge execution model.

2.1 5G Networks

Until 5th Generation (5G), each successive generation of mobile networks delivered incremental gains in throughput and reach while preserving the same fundamental paradigm: a purpose-built, hardware-centric network optimized for a single traffic type. 5G was designed from the ground up as a programmable, multi-service platform capable of simultaneously serving heterogeneous vertical industries with vastly different performance requirements, instead of simply aiming to increase throughput and extend data rates [1].

To address these diverse requirements, the ITU-R IMT-2020 framework defined three canonical usage scenarios: Enhanced Mobile Broadband (eMBB), targeting peak data rates in the tens of gigabits per second; Ultra-Reliable Low-Latency Communications (URLLC), targeting sub-millisecond latency and near-perfect reliability for mission-critical applications; and massive Machine-Type Communications (mMTC), supporting connection densities of up to one million devices per square kilometer for Internet of Things (IoT) deployments [2]. The 5G system achieves this flexibility through the adoption of key enabling technologies such as NFV, SDN, Network Slicing, and MEC, that transform the network into a software-defined, service-based, distributed infrastructure [3], [4].

Figure 2.1 shows the reference architecture for the service-based architecture of the 5G core, where each function exposes REST-based Application Programming Interfaces (APIs) over the Service-Based Interface. The upper functions belong to the control plane, while the bottom functions relate to the user plane.

The most relevant network functions of the 5G core are briefly described as follows [5]:

- The User Plane Function (UPF) is responsible for user plane packet handling such as routing/forwarding, marking and Quality of Service (QoS) enforcing;

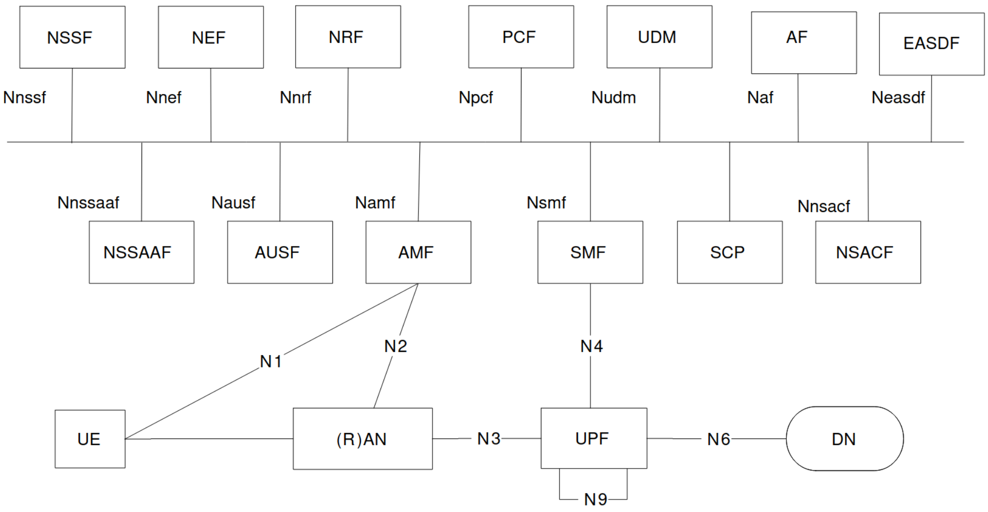

Figure 2.1: 5G Core Service Based Architecture [5]

- The Access and Mobility Management Function (AMF) manages registration, connection, mobility management and authorization and authentication;
- The Session Management Function (SMF) manages the establishment, modification, and termination of data sessions by configuring the UPF;
- The Policy Control Function (PCF) defines and enforces rules for QoS, charging and access control;
- The Network Exposure Function (NEF) exposes the capabilities and events of the 5G core to external applications and third parties in a controlled, authorized way;
- The Network Repository Function (NRF) allows network functions to register and discover each other, acting as a central service registry and discovery point for the Service Based Architecture;
- The Unified Data Management (UDM) stores and manages subscriber data and handles authentication credential generation and access authorization;
- The Authentication Server Function (AUSF) performs the actual authentication of the User Equipments (UEs) using the credentials from the UDM;
- The Network Slice Selection Function (NSSF) determines which network slice instances should serve the UE based on the set SLAs based on subscription info, policies, and requested slice types;
- The Network Slice-Specific Authentication and Authorization Function (NSSAAF) performs authentication and authorization of UEs specific to a network slice, enabling per-slice access control beyond the primary authentication;
- The Application Function (AF) represents external or operator applications which are able to interact with the 5G core via the NEF.

Complementing the core, the Radio Access Network (RAN) is responsible for the radio interface between UEs and the network, handling wireless transmission, scheduling, and handover. In the 5G architecture, the RAN is realized by Next Generation Node Bs (gNBs), which connect to the core via the NG interface and to one another via the Xn interface for mobility coordination [6].

2.1.1 5G-Advanced and the 6G Vision

Starting with Release 18 (Rel-18) [7] and continuing through Release 19 (Rel-19) [8], the 3rd Generation Partnership Project (3GPP) standardization roadmap extended the 5G baseline through a series of incremental releases, collectively referred to as 5G-Advanced. These releases introduce native Artificial Intelligence (AI) and Machine Learning (ML) capabilities across both the RAN and core network, enabling data-driven optimization of resource management, handover decisions, and load balancing. Further enhancements address cross-domain network slicing, extended support for XR and immersive services, reduced-capability device classes targeting Industrial Internet of Things (IIoT) and wearables, and significant improvements in energy efficiency. The cross-domain slicing enhancements in particular extend slice-aware MEC resource management beyond single-operator boundaries, establishing the groundwork for the multi-domain federation requirements being formalized for the sixth generation under the ITU-R IMT-2030 framework. The ITU-R IMT-2030 standardization process is expected to conclude around 2027, with 3GPP targeting the first native 6G releases (Release 20 and beyond) for commercial availability circa 2030 [9].

The ITU-R defines six usage scenarios for IMT-2030, expanding upon the three scenarios introduced in IMT-2020 while introducing new scenarios that previous generations were not designed to support. To support these scenarios, the IMT-2030 framework specifies substantially elevated technical performance requirements relative to IMT-2020: a peak data rate of 1 Tbps, an air-interface latency target of 0.1 ms, a connection density of 10^7 devices per square kilometer, a hundredfold improvement in energy efficiency, and sub-metre positioning accuracy [9]. The six usage scenarios are defined as follows:

- *Immersive Communication*: Extends eMBB to cover high-bandwidth, low-latency interactive experiences, including XR, holographic telepresence, and multi-sensory communications;
- *Hyper Reliable and Low-Latency Communication*: Extends URLLC to address mission-critical use cases where failure to meet reliability and latency requirements carries severe consequences, such as industrial automation, emergency services, and telemedicine;
- *Massive Communication*: Extends mMTC to support the connection of a massive number of devices and sensors across domains such as smart cities, logistics, agriculture, and environmental monitoring.
- *Ubiquitous Connectivity*: Addresses the digital divide by extending meaningful coverage to rural, remote, and underserved areas through interworking with non-terrestrial networks and other access systems.

- *Artificial Intelligence and Communication*: Integrates distributed AI workloads, including model training, inference, and computing resource orchestration, natively into the network infrastructure, enabling use cases such as autonomous driving assistance and digital twin creation;
- *Integrated Sensing and Communication*: Extends the radio interface beyond connectivity to provide sensing capabilities, including positioning, object detection, environmental monitoring, and real-time mapping, without requiring dedicated sensor hardware.

Beyond expanded scenarios and tightened performance targets, IMT-2030 envisions a structural departure from the AI-augmented design of 5G-Advanced toward an AI-native architecture, in which machine learning is an intrinsic design principle at every network layer rather than an overlay optimization mechanism [9]. Non-Terrestrial Network (NTN) integration, encompassing low Earth orbit satellite constellations and high-altitude platform systems, is elevated from a coverage supplement to a first-class access tier. Reconfigurable Intelligent Surfaces (RISs), which enable programmable reconfiguration of radio propagation environments without active signal generation, introduce a new infrastructure primitive for coverage and interference management. Network digital twins, which are software replicas of the physical infrastructure maintained in real time from live telemetry, are expected to serve as native management primitives, enabling closed-loop automation, predictive resource allocation, and intent verification prior to deployment.

These scenarios and architectural imperatives do not stand apart from the enabling technologies surveyed in this chapter; they impose new requirements on each of them. Particularly, Ubiquitous Connectivity at 6G scale requires seamless service continuity across operator and administrative boundaries, which in turn demands the programmable, runtime-negotiated federation mechanisms addressed in the multi-domain federation context of this work. Together, these requirements frame the technologies reviewed in the next chapters not as isolated background technologies but as co-dependent enablers of the 6G vision.

2.2 Network Functions Virtualization

Traditional network architectures relied heavily on proprietary hardware appliances for each specific function, such as firewalls, routers, and load balancers. These hardware-based solutions are not only costly but also inflexible, making it challenging to scale and adapt to dynamic network demands [10].

NFV allowed the transition from hardware-dependent network functions to virtualized software solutions. By leveraging virtualization techniques, NFV allows for dynamic deployment and lifecycle management of various network functions on Commercial Off-The-Shelf (COTS) hardware as Virtual Network Functions (VNFs). This approach decouples the functions of the network from proprietary hardware, enabling them to run on standard servers, storage, and networking equipment.

Virtualization brings several advantages over hardware-centric approaches [11]. VNFs run on COTS hardware, eliminating dependency on proprietary appliances and reducing both capital and operational expenditure. Functions can be deployed, scaled, and relocated dynamically across geographically distributed sites, enabling a degree of flexibility that static hardware cannot match. Standardized, open interfaces promote a multi-vendor ecosystem where VNFs and infrastructure components from different providers can coexist, and the rapid instantiation of new services reduces time-to-market significantly.

In the recent years, the dominant virtualization technologies have since shifted from hypervisor-managed virtual machines to containers orchestrated by platforms such as Docker[1] and Kubernetes[2], giving rise to Cloud-native Network Functions (CNFs): network functions decomposed into microservices and managed through cloud-native orchestration primitives. Container-based deployments offer lower per-instance overhead, faster startup times, and compatibility with Continuous Integration/Continuous Deployment (CI/CD) deployment workflows [12].

Figure 2.2 presents the ETSI NFV reference architecture, which organises the framework into three main domains: the VNFs, the NFV Infrastructure (NFVI), and the Management and Orchestration (MANO). VNFs are software implementations of network functions that execute on NFVI resources; the NFVI encompasses the compute, storage, and networking hardware and software that host them. The MANO domain coordinates the lifecycle of both VNFs and NFVI resources, and is itself subdivided into three building blocks:

- *Virtual Network Functions Managers (VNFMs)*: responsible for managing the lifecycle of VNFs, including instantiation, scaling, updating, and termination. It interacts with both the Virtual Infrastructure Manager (VIM) and NFV Orchestrator (NFVO) to ensure the VNF operates as intended within the NFV infrastructure.
- *VIMs*: manages the NFV infrastructure resources, such as compute, storage, and networking, within a single administrative domain. It provides resource abstraction and allocation, enabling the NFVO and VNFM to deploy and manage VNFs.
- *NFVOs*: is responsible for orchestrating and managing network functions across multiple VIM instances and ensuring the overall coordination of resources and services within the NFV environment. It also handles policy-based resource optimization and SLA management.

The other components of the framework are the Operation Support System (OSS)/Business Support System (BSS), which allows for NFV service providers to interact with the NFV infrastructure, enabling service provisioning, billing, and customer management; the Service, VNF and Infrastructure Descriptions, which contain templates/descriptors used internally within the NFVO to be translated into functional blocks; and the Element Management System (EMS) which manages the functionality of one or more VNFs.

[1]https://www.docker.com/
[2]https://kubernetes.io/

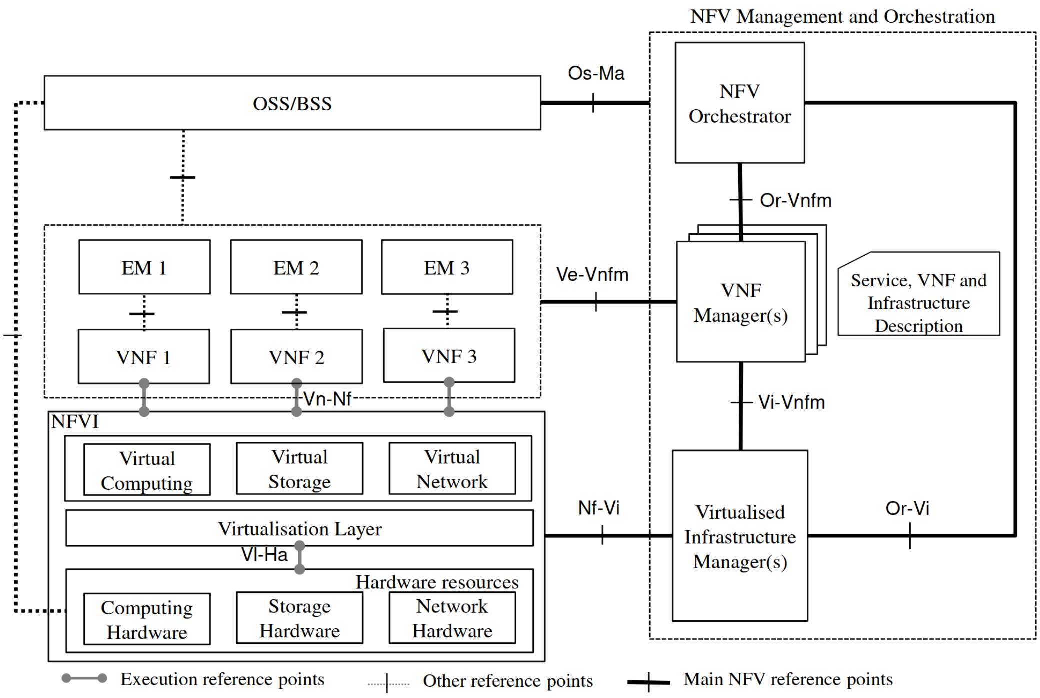

Figure 2.2: ETSI NFV reference architecture [11]

2.3 Software Defined Networking

In conventional network architectures, routing decisions, forwarding table computation, and policy enforcement are embedded within each device and managed independently through vendor-specific interfaces. With no global view of network state and per-device configuration as the only operational primitive, adapting to traffic demands or failures is slow and costly [13].

SDN addresses these limitations by decoupling the control plane, responsible for computing forwarding decisions and enforcing network-wide policies, from the data plane, responsible for packet forwarding [14]. Network intelligence is consolidated in a logically centralized controller, which maintains a global view of the network topology and programs forwarding behavior across the infrastructure through standardized interfaces. This separation transforms the network into a programmable, software-driven resource that can be dynamically configured to meet changing service demands. By holding a consistent model of the full topology, the controller enables network-wide optimizations, such as coordinated policy enforcement, real-time flow rerouting, and automated provisioning, that are impractical when control is fragmented across independent devices. Furthermore, standardized southbound interfaces allow the controller to manage forwarding elements from heterogeneous vendors, and northbound APIs expose network resources to higher-level orchestration systems, enabling tight coordination with complementary frameworks such as NFV.

Beyond these core properties, SDN enables [14]:

- *Vendor Agnosticism*: Standardized southbound interfaces reduce dependency on proprietary hardware and vendor lock-in;
- *Dynamic Traffic Management*: The controller can reroute flows in real time based on

current conditions, application requirements, or operator-defined policies;

- *Network Automation*: Programmable APIs support automated lifecycle management and coordination across the infrastructure.

Figure 2.3 shows an overview of the SDN architecture, which is organized into three functional layers, each communicating through standardized interfaces.

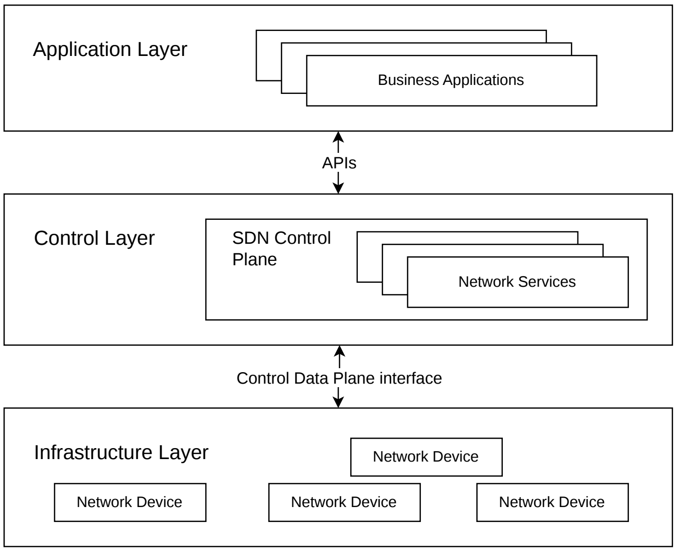

Figure 2.3: SDN three-layer architecture (adapted from [13])

The infrastructure layer comprises the physical and virtual forwarding devices that constitute the data plane. These devices do not compute forwarding decisions autonomously; instead, they execute flow rules programmed by the controller, matching incoming packets against installed entries and applying the associated forwarding, dropping, or marking actions defined in their flow tables.

The control layer hosts the SDN controller, which maintains a logically centralized, real-time model of the network topology and computes forwarding policies on behalf of the infrastructure devices. The controller translates high-level intents into low-level flow rules and disseminates these rules to the appropriate forwarding elements.

The application layer contains network applications that consume the controller's abstractions to implement higher-order network behaviors, such as load balancing, traffic engineering, access control, and network slice management. Applications interact with the controller without requiring knowledge of the underlying hardware topology or forwarding mechanisms.

These layers communicate through three categories of interface. The southbound interface connects the controller to infrastructure devices; OpenFlow is the dominant protocol, defining the message format and procedure for installing, modifying, and querying flow tables on forwarding elements [15]. The northbound interface exposes the controller's network model and programming capabilities to applications, typically through REST APIs or intent-based abstractions; no single standard governs this interface, and implementations vary across controller platforms. An east-west interface can also be considered, enabling coordination between multiple controller instances and supporting distributed or hierarchical controller deployments where no single instance maintains a complete view of a large-scale network.

Within the 5G architecture, SDN principles are directly reflected in the design of the UPF, whose forwarding behavior is programmatically controlled by the SMF through the N4 interface. This design applies the same control/data plane separation that defines SDN natively within the 5G core, enabling per-session traffic steering, QoS enforcement, and data path programmability without relying on static, hardware-bound forwarding configurations [5].

2.4 Network Slicing

The heterogeneous nature of 5G use cases imposes a fundamental tension on network design: the performance requirements of eMBB, URLLC, and mMTC services are not only distinct but often irreconcilable within a single, monolithic network configuration. A configuration tuned for peak throughput necessarily compromises the deterministic latency and reliability demanded by mission-critical applications, while one hardened for ultra-reliability wastes resources when serving bandwidth-intensive media services. Network slicing addresses this constraint by enabling the concurrent operation of multiple logically isolated, independently configured network instances over a shared physical infrastructure, each optimized for the characteristics of its target service [16].

A network slice is an end-to-end logical network instance comprising a set of network functions and associated resources (compute, storage, and transport) configured and isolated to satisfy the specific Key Performance Indicators (KPIs) of a given service. Each slice spans end to end, from the RAN through the transport domain to the 5G core, and is defined by a Network Slice Template (NST), which captures the performance targets, resource budgets, and topology constraints that govern instantiation and operation. Slices are isolated from one another across the data, control, and management planes, ensuring that faults and congestion within one slice cannot propagate to others.

From an operational standpoint, slicing yields several properties that a monolithic design cannot provide [5]:

- *Logical Isolation*: Each slice operates as an independent network instance, ensuring that traffic, faults, and resource contention within one slice do not affect others;
- *Customized QoS*: Slice-specific configurations enable delivery of differentiated service profiles tailored to each vertical, from high-bandwidth media delivery to ultra-reliable industrial control;
- *Multi-tenancy*: Multiple operators, enterprises, or service providers can share the same physical infrastructure while receiving logically dedicated resources and isolated management planes;
- *Rapid Service Deployment*: Slices can be instantiated, modified, and decommissioned programmatically through standardized interfaces, reducing the time-to-market for new services;

- *Efficient Resource Utilization*: Resources can be dynamically allocated and reallocated across slices according to demand, improving overall infrastructure efficiency compared to statically partitioned, purpose-built networks.

Network slicing is not an independent abstraction but is technically realized through the joint application of NFV and SDN. NFV provides the virtualization layer that enables per-slice network function instances to be created, configured, and terminated on shared COTS hardware. Without NFV, deploying independent function sets per slice would require dedicated physical appliances for each service type, negating the shared-infrastructure principle that slicing depends upon. The NFVO and VNFM components of the MANO framework orchestrate the lifecycle of these per-slice VNFs, while the VIM allocates and tracks the compute, storage, and networking resources budgeted to each slice instance.

SDN complements this by enforcing per-slice traffic isolation and steering in the data plane. The SDN controller programs forwarding rules that segregate slice-specific flows across the shared physical underlay, and enforces policies that prevent cross-slice interference at the transport layer. This mechanism extends the control/data plane separation already present in the 5G core (where the SMF programs the UPF via the N4 interface) to an end-to-end scope spanning the RAN, transport, and core simultaneously.

Together, NFV governs the function plane (instantiation, scaling, and termination of per-slice network functions) while SDN governs the transport plane (isolation, steering, and policing of per-slice flows). The slice management layer coordinates both through MANO APIs and SDN controller northbound interfaces, translating high-level service requirements into concrete resource allocations and forwarding policies across the full network stack.

The management of network slices in the 3GPP architecture, as defined in TS 23.501 [5], is organized into three hierarchical functional layers, as illustrated in Figure 2.4:

- *Communication Service Management Function (CSMF)*: translates high-level customer-facing service requirements into network slice requirements, acting as the interface between business intent and the network management plane;

- *Network Slice Management Function (NSMF)*: manages the lifecycle of Network Slice Instances (NSIs), coordinating the instantiation and operation of constituent sub-slices and enforcing end-to-end SLAs;

- *Network Slice Subnet Management Function (NSSMF)*: manages the lifecycle of individual Network Slice Subnet Instances (NSSIs) within RAN, transport and core, allocating and configuring the specific resources required by each sub-slice.

Network slices are identified through the Network Slice Selection Assistance Information (NSSAI), a collection of one or more Single Network Slice Selection Assistance Informations (S-NSSAIs). Each S-NSSAI comprises two fields: the Slice/Service Type (SST), which encodes the service category, and an optional Slice Differentiator (SD), which allows operators to differentiate among multiple slices of the same service type. The 3GPP specification standardizes four SST values: eMBB (1), URLLC (2), mMTC (3), and Vehicle-to-Everything (V2X) (4). The 24-bit SD field provides an operator-defined namespace for further differentiation.

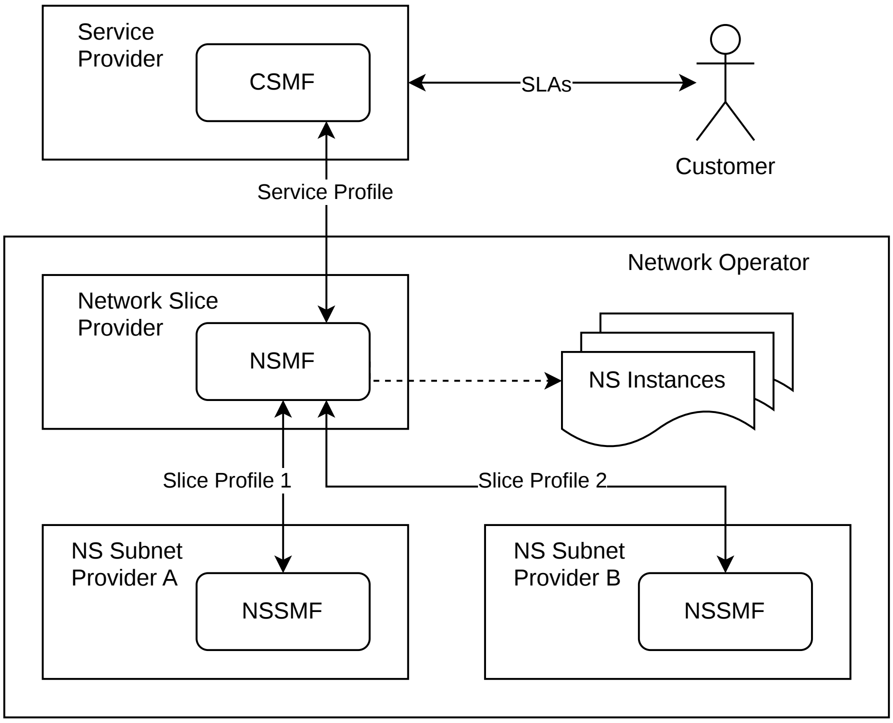

Figure 2.4: Network Slicing Management Functions (adapted from [17])

Slice selection is handled at registration time. The UE includes a requested NSSAI in its registration request, which the AMF forwards to the NSSF. The NSSF evaluates the request against the UE's subscription profile, operator policy, and current resource availability, returning an allowed NSSAI and, where necessary, a redirection to an AMF set associated with the target NSI.

An end-to-end NSI is composed of multiple NSSIs, one per network domain - RAN, transport and core. The RAN sub-slice configures the radio resources and RAN functions assigned to the slice; the transport sub-slice establishes the connectivity between the RAN and the core; the core sub-slice instantiates the relevant 5G core network functions with the appropriate configurations and policies. Sub-slices may be shared across multiple NSIs where isolation requirements permit, enabling more efficient use of underlying resources.

The lifecycle of a network slice spans five phases: preparation, in which the NST is onboarded and feasibility is verified; instantiation, in which the constituent NSSIs are created and configured; activation, in which the slice begins carrying live traffic; operation, in which the slice is continuously monitored and dynamically adapted to changing demand; and decommissioning, in which the slice is gracefully terminated and its resources released. Lifecycle management is coordinated between the NSMF and the domain-specific NSSMF instances, following the interface specifications defined in 3GPP TS 28.530 [18].

2.5 Multi-access Edge Computing

Latency-sensitive applications, such as real-time video analytics, augmented reality, industrial automation, and autonomous vehicle coordination, expose a structural limitation of the centralized cloud model. By concentrating compute and storage in geographically distant data centers, the cloud introduces round-trip latencies on the order of tens of milliseconds, even under ideal network conditions. For applications that require deterministic, sub-millisecond response at the point of access, this is not merely a performance constraint but a functional barrier [19].

MEC, standardized by ETSI [20], resolves this by extending cloud-computing capabilities to the edge of the RAN, co-locating compute, storage, and networking resources in direct proximity to the UE. This placement eliminates the backhaul round-trip to the core, enabling service execution at or near the base station. By exposing these resources to authorized applications through standardized APIs, MEC transforms the network edge from a passive radio access point into an active, programmable service execution environment. These properties are impractical to achieve in a centralized deployment:

- *Ultra-Low Latency*: Application execution at the network edge eliminates the round-trip to centralized data centers, enabling response times compatible with real-time and mission-critical use cases;
- *Backhaul Offload*: Local processing of traffic at the edge reduces upstream bandwidth consumption, alleviating congestion on backhaul links and improving overall network efficiency;
- *Data Locality*: Processing data in proximity to its source allows operators and enterprises to enforce data residency and sovereignty requirements without routing sensitive information to remote infrastructure;
- *Application Mobility*: MEC supports the live relocation of application instances across hosts in response to UE mobility, preserving service continuity without interruption.

The reference architecture for MEC, as defined by ETSI in GS MEC 003 [20], is presented in Figure 2.5. The architecture is organized into two principal domains: the MEC Host (MEH), which hosts applications and exposes platform services, and the MEC Management system, which governs the lifecycle of applications and the allocation of resources across the deployment.

The MEH comprises two co-located components: the MEC Platform (MEP) and the Virtual Infrastructure (VI). The VI provides the compute, storage, and networking resources on which MEC applications execute, and includes a data plane responsible for enforcing traffic rules and routing flows among applications, services, and external networks. The MEP operates above the VI, providing the service environment through which applications discover and consume MEC services, register their resource requirements, and interact with the broader network. Specifically, the MEP manages traffic rules, configures Domain Name System (DNS), and hosts platform-level services such as location and radio network information.

MEC applications execute as virtualized entities (containers or virtual machines) within the MEH. Their interaction with the MEP and with one another is governed by three reference points: Mp1 connects applications to the MEP, enabling service registration and discovery,

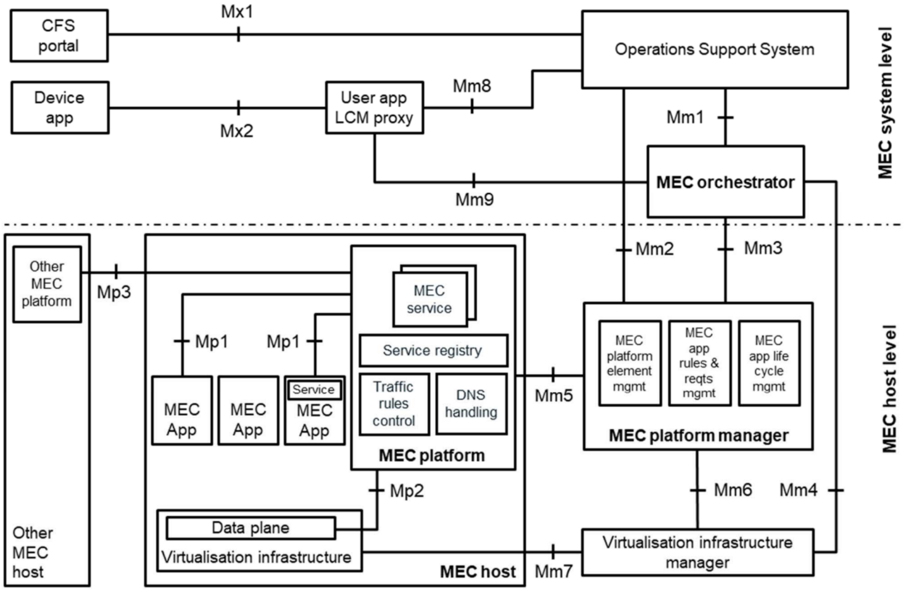

Figure 2.5: ETSI MEC reference architecture [20]

traffic rule activation, and session state relocation in response to mobility events; Mp2 connects the MEP to the VI data plane, allowing the platform to steer traffic flows among applications, services, and external interfaces; Mp3 enables inter-platform communication between MEP instances on separate hosts, supporting coordinated service delivery across distributed deployments.

The MEC Management system is organized into two tiers. At the system level, the MEC Orchestrator (MEO) maintains a global view of deployed hosts, available resources, and accessible services, and is responsible for the end-to-end lifecycle of MEC applications, including onboarding, instantiation, termination, and application relocation decisions across hosts. At the host level, the MEC Platform Manager (MEPM) manages the operational lifecycle of applications on a single host, handling rule enforcement, conflict resolution, and DNS configuration, while the VIM allocates, monitors, and releases the virtualized resources required by those applications, reporting performance metrics and fault conditions to the MEPM.

These management functions interact through a set of Mm reference points. Mm1 connects the MEO to the operator's OSS, enabling application instantiation and termination requests to be submitted from the operational layer. Mm3 links the MEO to the MEPM for application lifecycle coordination and service management. Mm4 connects the MEO to the VIM for resource allocation and tracking at the host level. Mm5 links the MEPM to the MEP, supporting platform configuration, lifecycle support, and application relocation procedures. Mm2, Mm6, and Mm7 extend these interactions between the OSS, MEPM, VIM, and VI for configuration, fault management, and performance monitoring. The OSS bridges the operational and business layers of the system, managing service provisioning, billing, and customer management, and receiving service delivery requests from the Customer Facing System (CFS) via the Mm8 reference point.

2.5.1 MEC Integration with NFV, SDN, and Network Slicing

MEC does not operate in isolation from the enabling technologies described in the preceding sections; rather, it builds directly on NFV, SDN, and network slicing to deliver its edge execution model [21].

The VI within each MEH is, in architectural terms, an instance of the NFV infrastructure deployed at the network edge. MEC applications execute as virtualized entities, such as virtual machines or containers, managed by a VIM that is functionally equivalent to its NFV-MANO counterpart. The lifecycle procedures carried out by the MEPM and MEO, like application onboarding, instantiation, scaling, relocation, and termination, directly mirror the VNFM and NFVO procedures defined by the ETSI NFV framework, with the addition of mobility-driven relocation as an edge-specific operational requirement.

SDN underpins the traffic steering that directs flows from UEs or from the core network to the appropriate MEC application. The Mp2 reference point between the MEP and the VI data plane functions as a southbound control interface: the MEP installs traffic rules that redirect specific flows to registered application endpoints on the host. At a larger scale, SDN controllers operating at the transport layer can steer UE traffic toward the nearest or most appropriate MEC host as users move across the coverage area, enabling network-wide traffic management that complements the host-local rule enforcement provided by Mp2.

Network slicing extends MEC into a multi-tenant edge execution environment. A single MEH can host applications belonging to multiple concurrent slices, with compute, storage, and networking resources partitioned according to per-slice SLAs. The NSSMF, acting on behalf of the NSMF, can coordinate with the MEO and MEPM to instantiate slice-specific edge applications as components of the core or RAN sub-slice, ensuring that edge resource allocation is governed consistently with the end-to-end performance contract of the slice. This integration, sometimes referred to as slice-aware MEC, is the architectural basis for deploying differentiated edge services on the same physical host while preserving mutual isolation.

2.6 Intent-Based Service Orchestration

As network infrastructure has grown in complexity, the operational burden of managing these systems through traditional low-level, vendor-specific interfaces has become a significant constraint. Operators must translate business-level intents, such as provisioning a specific slice or offloading edge workloads, into sequences of precise API calls across heterogeneous management stacks. This translation requires deep technology expertise, is error-prone, and scales poorly as the number of managed systems and administrative domains grows.

The emergence of Large Language Models (LLMs) capable of reasoning over natural language and structured API specifications has opened a new class of solutions: management portals that accept operator intents expressed in natural language and autonomously generate and execute the corresponding API call sequences. This approach, broadly referred to as AI-augmented or intent-based network management, is increasingly being adopted by both standardization bodies and open-source platform initiatives.

LLMs, pre-trained on large corpora of text and code, exhibit strong capabilities in tasks directly relevant to network management: understanding technical specifications, generating configuration templates, reasoning over API semantics, and translating high-level objectives into structured call sequences [22]. These capabilities enable portals that accept natural-language descriptions of desired network states and produce the API invocations required to achieve them. Standardization bodies have begun incorporating AI into management frameworks: 3GPP Rel-18 introduces native AI/ML support across both the RAN and core network management planes, while the GSMA and ETSI have each initiated work on AI-assisted capability exposure and operator-facing portals.

Intent-based network management captures the paradigm in which operators express desired network behaviors or service outcomes as high-level intents, rather than specifying the individual configuration steps required to achieve them [23]. Intents may range from provisioning a URLLC slice for an enterprise customer to offloading edge workloads to a federated partner during peak demand; the management system is responsible for decomposing, translating, and executing each intent against the underlying infrastructure. Earlier approaches relied on formal intent languages and YANG-based models, which require operators to learn specialized syntax and constrain expressiveness to what the formal model can encode. LLMs offer an alternative: by leveraging their understanding of natural language and API semantics, they can accept intents expressed in plain language and perform decomposition and translation automatically, making intent-based management accessible without specialized training.

LLMs alone cannot invoke external APIs: they generate text but have no execution mechanism to call live systems. The Model Context Protocol (MCP)[3] is an open protocol that defines a standardized interface through which LLMs can discover and invoke external tools, each described by a name, a natural-language description, and an input schema. In the context of network management, an MCP server can expose telecom management APIs as tools available to the LLM, cleanly separating natural-language reasoning from API execution. This decoupling allows the same LLM to target heterogeneous management APIs by swapping or extending the server's tool definitions.

An LLM agent extends single-shot inference into an autonomous perception–reasoning–action loop: the model observes the current environment state through tool outputs, reasons about the next action, executes that action via an MCP tool call, observes the result, and iterates until a goal condition is satisfied [24]. The distinction from plain LLM inference is not the underlying model but the control loop: repeated model calls are driven by an external harness that routes tool outputs back as subsequent inputs, transforming the LLM from a one-shot translator into a goal-directed autonomous executor. This makes LLM agents particularly suited to network orchestration tasks that are inherently multi-step and stateful: deploying a network slice requires not only generating the correct API call sequence but also verifying each intermediate state transition, handling partial failures, and adapting when conditions change.

[3] https://modelcontextprotocol.io/

Multi-agent systems extend the agent model further by composing multiple LLM-backed agents, each scoped to a distinct responsibility boundary and equipped with its own tool set, under a coordinating orchestrator agent that decomposes high-level intents, delegates subtasks, and aggregates results. Each agent maintains its own context and acts autonomously within its scope, enabling concurrent execution of independent subtasks and clean separation of concerns across functional or administrative boundaries.

2.7 Multi-domain Network Federation

Multi-domain network federation refers to the voluntary sharing of resources, capabilities, and services across distinct administrative domains under agreements that preserve each domain's operational autonomy and policy sovereignty [25]. Unlike roaming, which has historically been mediated by static, bilaterally negotiated arrangements for user-plane connectivity, federation in the context of 5G and MEC is dynamic and programmable: operators negotiate and activate cross-domain resource commitments at runtime, through standardized interfaces, in response to observed demand.

The fundamental challenge of federation is coordinating across administrative trust boundaries without compromising domain sovereignty. Each operator controls its own MANO stack, resource allocation policies, and security perimeter. A federation framework must therefore define inter-domain interfaces through which capability advertisement, resource negotiation, and lifecycle coordination can take place without requiring either domain to expose its internal topology or proprietary configurations to the other. This separation between what is shared and what remains internal is enforced at the federation boundary, typically through a dedicated federation gateway or brokerage layer.

Federation interactions can follow several patterns, depending on the operational scenario:

- *Resource Brokerage*: A consumer domain acquires computational or network capacity from a provider domain to augment its own, with the commitment agnostic to any specific workload;
- *Service Chaining*: An end-to-end service or network slice spans multiple domains, with each contributing a segment; the federation layer stitches these segments into a coherent, SLA-compliant service;
- *Workload Offloading*: A domain experiencing transient resource saturation redirects application instances or traffic flows to a federated partner, maintaining service continuity through dynamic load redistribution.

2.7.1 GSMA Operator Platform Group

The GSMA Operator Platform initiative, developed by the Operator Platform Group (OPG), defines a framework through which mobile operators can expose their network and edge capabilities to application developers, enterprises, and peer operators through standardized, programmable interfaces [26]. Three principal objectives guide the initiative:

enabling vendor-agnostic capability exposure by abstracting over heterogeneous internal infrastructure; supporting programmable federation between peer operators without requiring pre-negotiated bilateral arrangements; and providing a unified API surface through which application developers and enterprises can consume edge and network resources across multiple operators without per-operator integration effort.

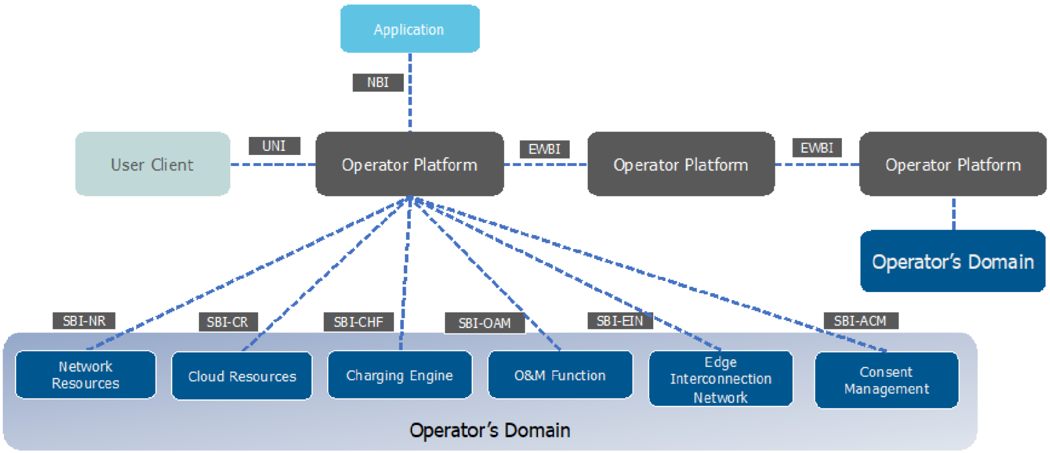

Figure 2.6: GSMA OP reference architecture [26]

The Operator Platform (OP) reference architecture is illustrated in Figure 2.6. The OP's functionality is structured around three distinct roles, each addressing specific aspects of service management. These are the Capabilities Exposure role, the Service Resource Manager Role, and the Federation Manager role.

The Capabilities Exposure role exposes the capabilities of the OP via its North Bound Interface (NBI), facilitating discovery of functionality (e.g. MEC application/network slice onboarding and Life Cycle Management (LCM)) and operational capabilities (e.g., deployment locations and QoS attributes). It is responsible for enabling an operator, application provider or consumer to operate their services provided by the OP.

The Service Resource Manager role manages the operator's network infrastructure resources via the South Bound Interface (SBI) and the User Network Interface (UNI). The SBI incorporates multiple interfaces that connect the OP to other network functions and enablers required to execute the services exposed by the OP. On the other hand, the UNI allows a user client to interact with the OP to access services and resources.

The Federation Manager role is responsible for interacting with other OPs to establish federation relationships and exploit them via the EWBI. It allows an OP to establish and sustain federation, share and update resource catalogues including details about available infrastructure resources, and facilitates resource reservation and service deployment in partner networks while respecting federation policies and operational agreements.

2.7.2 ETSI MEC Federation and OpenOP

As the deployment of MEC spread across operator networks, the single-domain scope of the original GS MEC 003 reference architecture proved insufficient for scenarios requiring cross-operator service continuity. To address this gap, ETSI introduced a new MEC reference architecture variant for MEC Federation (MECF) support, shown in Figure 2.7, which includes the new MEC Federator (MEF) component, along with two reference points: the Mfm for MEF-to-MEC Application Orchestrator (MEAO) communication, and the Mff for MEF-to-MEF signaling. Furthermore, the Group Specification (GS) MEC 040 [27] was published, starting the specification for the Federation Enablement APIs of the Mff interface. Other relevant documents such as Group Report (GR) MEC 035 [28], which establishes the scope and use cases of the MECF framework, and ETSI White Paper 49 which offers deployment guidance, establishes terminology and classifies architectural option for federated MEC deployments, provide more additional insights on the edge federation problem.

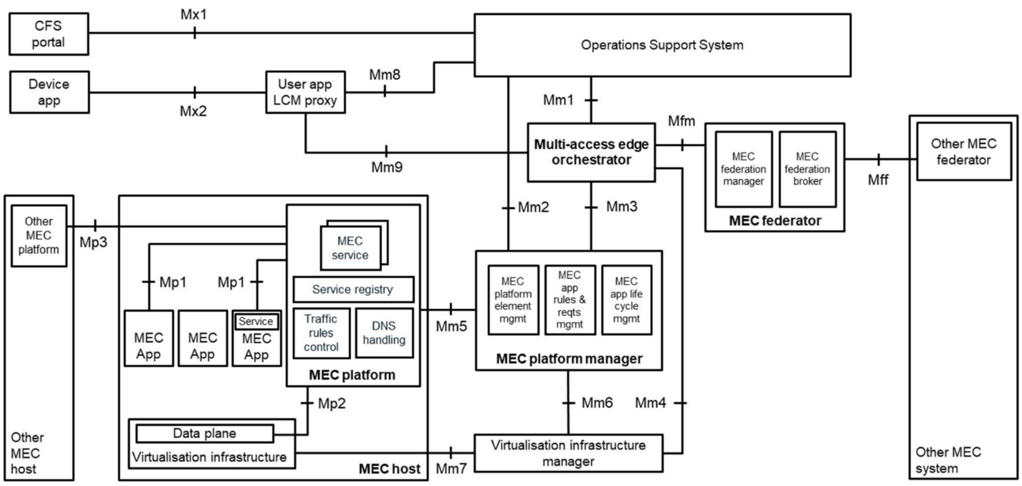

Figure 2.7: ETSI MEC federation architecture variant [27]

Currently, the normative work of ETSI, GSMA, and other industry bodies such as Connected APIs for Mobile and Networking (CAMARA) [29] have produced coherent specifications, but operators and application developers still lack a vendor-neutral, deployable reference implementation of the against which to validate interoperability. This fact led to the establishment of a new Software Development Group (SDG) at ETSI, the SDG Open Operator Platform (OpenOP). Its aim is to provide an open-source reference implementation of the GSMA OPG, that can then serve as an experimentation platform for 6G testbed federation and telco cloud scenarios and help generate implementation feedback to inform future standardization cycles [30].

Figure 2.8 shows the architecture for the first integrated release of the OpenOP, which closely relates to the OP reference architecture. The Open Exposure Gateway functions as the northbound API gateway, exposing telecom capabilities and edge cloud resources through CAMARA APIs; it maps directly to the NBI of the GSMA OP reference architecture and provides the interface through which application developers and enterprises consume edge resources across participating operators. The Service Resource Manager handles application lifecycle orchestration and coordinates deployment across edge cloud zones by integrating with underlying NFV and MEC MANO stacks; it implements the core resource controller and service registry functions of the GSMA OP architecture and mediates the SBI toward heterogeneous operator infrastructure. The Artificial Intelligence Integration (AI^2) module provides intent-based interaction through an MCP server backed by an LLM, translating natural-language operator intents into CAMARA API calls. The Federation Manager implements GSMA's EWBI API specifications in order to enable federation between distinct domains.

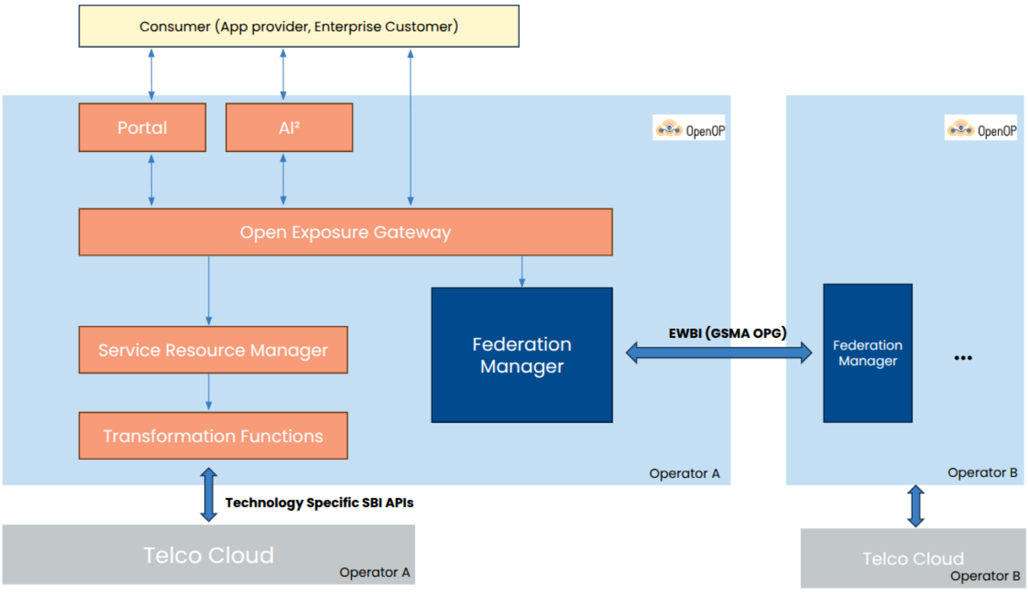

Figure 2.8: ETSI OpenOP Release 1 architecture [30]

Building on the Release 1 foundation, a set of incremental improvements targeting architectural maturation are underway as part of Release 1.1. First, the NBI will be extended to support additional standard API types beyond CAMARA, incorporating TM Forum Open APIs and GSMA to broaden interoperability with existing OSS/BSS ecosystems and align with operator onboarding workflows that already rely on TM Forum interfaces. Second, the internal component architecture will be refactored toward greater abstraction, as the Release 1 design coupled internal logic too tightly to standard-specific constructs; the refactoring decouples components from standard assumptions and re-orients them around infrastructure primitives, enabling interoperability with heterogeneous MANO deployments while still focusing on standard-driven exposure. Third, a data bus will be introduced as a horizontal messaging layer, enabling both synchronous request-response and asynchronous event-driven interactions across components; synchronous where latency-sensitive coordination is required, and asynchronous where decoupled notification or telemetry flows are more appropriate. A subsequent Release 2 is currently in the planning stage.

CHAPTER 3

Research Statement

The road towards the next generation of networks has accelerated the deployment of edge computing infrastructure and network slicing across heterogeneous, multi-operator environments. Vertical industries with fundamentally different requirements demand service levels that no single operator can guarantee in isolation. Existing federation standards, such as the ones provided by GSMA and ETSI, provide conceptual frameworks and initial API specifications, but a complete and comprehensive implementation of a federated, slice-aware management framework still does not exist. This proposal is primarily grounded in these normative specifications and industry-driven frameworks; the systematic literature review planned for Work Package (WP)1 will complement this perspective with a broader survey of the research literature on open challenges and existing solutions.

In this sense, we propose to research a multi-level federation architecture that integrates private edge infrastructure with public telecom clouds to address these challenges. The central contribution is a stateless MEC slice mobility framework that treats network slice components as portable, self-contained microservice bundles decoupled from any origin domain's internal state, enabling live relocation across administrative boundaries. Complementing this, the thesis specifies inter-domain orchestration mechanisms based on the EWBI of the GSMA Operator Platform, supporting dynamic capability advertisement, resource reservation, and SLA negotiation between domains with no prior relationship. We also plan to integrate intent-based interaction into the solution to reduce the operational complexity of activating these cross-domain workflows.

The envisioned outcome is a deployable, open-source federation stack that allows edge operators to federate dynamically, activating resource commitments in real time in response to observed demand and without manual pre-configuration. Two representative scenarios will be used to evaluate the design. In the Smart Stadium scenario, a private edge deployment must elastically overflow into public MEC infrastructure during crowd-driven load spikes. In the cross-border PPDR scenario, first-responder communications must persist across domain transitions under strict reliability constraints. Validation in both scenarios, combined with open-source release of the resulting components, is expected to produce actionable implementation feedback for ongoing standardization work, alongside scientific contributions to the research community.

3.1 Research Questions

The main research questions are as follows:

- Q1. How can network slice mobility be efficiently managed in federated environments by treating MEC slices as stateless microservices?

- Q2. What are the optimal cross-domain interconnection mechanisms to ensure seamless service chaining and guaranteed performance between private edges and public networks?

- Q3. How can multi-level federation interfaces (e.g., EWBI) be dynamically automated to support real-time resource negotiation during sudden high-density events or critical inter-domain PPDR operations?

3.2 Goals

The main goals to be achieved are to:

- G1. Design a multi-level federation architecture that seamlessly integrates private edge networks with public telecom clouds.

- G2. Develop a slice mobility management framework that utilizes stateless MEC microservices to improve operational flexibility in high-density environments, treating slice components as portable, containerised microservice bundles with no stateful coupling to the origin domain.

- G3. Implement automated inter-domain service orchestration (including G1 and G2 principles), leveraging the EWBI for dynamic capability advertisement and SLA negotiation between domains with no prior relationship.

- G4. Design and integrate an intent-driven interaction layer, mediated through LLM-backed MCP servers, spanning all framework components to reduce the operational complexity of activating workflows.

- G5. Validate the proposed federation model against real-world constraints in a Smart Stadium crowd management scenario and an inter-domain cross-border PPDR communication scenario, meeting defined latency and continuity thresholds for each.

3.3 Validation Scenarios

The two scenarios adopted in this proposal were selected because they represent complementary extremes of the federated edge problem space, stressing different subsystems of the proposed framework and together providing coverage across all stated research questions and goals.

The Smart Stadium scenario models a high-density commercial deployment in which a private edge installation, sized for average attendance loads, becomes insufficient during peak events. Crowd-driven surges in demand for services such as real-time video analytics and crowd management require rapid, elastic overflow of compute capacity into public MEC infrastructure. This scenario highlights the elastic nature of federation: the ability to detect demand growth, negotiate resources with an external domain under no prior relationship, and migrate or instantiate slice components in real time while meeting latency bounds.

The cross-border PPDR scenario models a persistent, mission-critical deployment in which first-responder communications must remain operational as personnel cross administrative domain boundaries. Unlike the stadium case, demand is not the stress factor, but reliability is. Slice components must be relocated across domains with hard continuity guarantees, policy enforcement must persist through transitions, and service degradation is not acceptable. In this case, resilience is the main validation driver, making it the primary vehicle for validating slice handover continuity, cross-domain policy enforcement persistence, and zero-degradation SLA compliance under hard operational constraints.

Although these two scenarios currently constitute the validation effort, the architecture is not to be designed around them. Should the validation context shift, whether due to evolving requirements, testbed availability, or emerging use cases, the framework will be able to accommodate new scenarios without requiring fundamental architectural revision.

3.4 Expected Results

The expected results of the proposed research are:

- R1. A scalable architecture for private/public edge federation.
- R2. An open-source implementation of a stateless MEC slice mobility manager.
- R3. Demonstration of the architecture in a stadium scenario ensuring real-time operational technology responsiveness.
- R4. Demonstration of the architecture in a PPDR scenario ensuring inter-domain operational mobility.
- R5. Scientific publications in top-tier telecommunications conferences and journals.

CHAPTER 4

Workplan and Schedule

This chapter presents the research workplan, structured into five WPs. Then, expected deliverables are outlined, research milestones are identified, and a Gantt chart of the workplan is presented.

4.1 WP1: Technological Assessment and State of the Art

WP1 establishes the research foundation through systematic survey and critical analysis of the state of the art in edge networking, MEC, network slicing, and multi-domain federation, identifying open challenges and limitations of existing solutions that motivate and constrain the proposed research.

4.1.1 Task 1.1 - Literature Review

This task conducts a systematic review of the scientific literature on MEC slice mobility, private/public edge integration, and multi-domain network federation. The review targets the identification of open research challenges that motivate the proposed research, such as trust bootstrapping between previously unknown administrative domains, SLA enforcement across operator boundaries, scalability under high resource-state churn, and zero-prior-agreement federation.

4.1.2 Task 1.2 - Standards, Technologies and Solutions

Building on the literature survey, this task performs a structured analysis of relevant normative standards and open-source implementations. Standards from the ETSI MEC Industry Specification Group (ISG), the GSMA OPG, and 3GPP network slicing specifications will be reviewed. Solutions such as ETSI OpenOP Release 1 are assessed for possible concrete baselines.

4.2 WP2: Architecture Design for Federated Edge and Slicing

WP2 aims to design an architectural framework that accomplishes the necessary goals. It is divided into the following tasks:

4.2.1 Task 2.1 - Initial Architecture Design

This task designs architecture solutions for both research problems: a stateless MEC slice mobility framework that treats network slices as portable microservice bundles enabling migration across edge infrastructure without stateful coupling to the origin domain, and an inter-domain service chain orchestration framework specifying mechanisms for capability advertisement, resource reservation, and SLA negotiation between previously unknown administrative domains. The designs should be based on the OP architecture, specifically the inter-operator APIs (EWBI) and the exposure model.

4.2.2 Task 2.2 - Integrated Final Architecture Design

This task integrates the two candidate designs from Task 2.1 into a single coherent architecture that jointly addresses slice mobility and inter-domain orchestration. The integration identifies shared components, common abstractions, and cross-cutting concerns to produce a consolidated architectural specification that resolves both problems within a single deployable framework.

4.3 WP3: Implementation of Federation and Orchestration Mechanisms

WP3 implements the designed frameworks as functional open-source software components. Initially, prototypes validating the feasibility of each mechanism will be developed, and later integrated into a unified framework. (R1, R2).

4.3.1 Task 3.1 - Initial Implementations and Proof of Concepts

This task focuses on the implementation of the individual architectural components specified in WP2 Task 2.1 as functional prototypes, validating and testing the slice mobility and inter-domain federation mechanisms before integration into a consolidated framework. These prototypes will use ETSI OpenOP as a baseline, due to it being the most complete implementation of an OP up to date.

4.3.2 Task 3.2 - Final Integrated Architecture Implementation

This task implements the unified architecture specified in WP2 Task 2.2 as a coherent end-to-end system. The PoC components from Task 3.1 are integrated and extended, resulting in a slice mobility framework capable of sharing and mobility of edge resources across multiple providers. The resulting federation stack is released as an open-source software contribution.

4.4 WP4: End-to-End Validation in Deployment Scenarios

WP4 validates the developed system through two representative real-world deployment scenarios to produce concrete, quantified demonstrations of operational feasibility and performance (R3, R4).

4.4.1 Task 4.1 - Stadium Scenario

This task integrates and tests the stateless MEC slicing and federation framework in a high-density Smart Stadium environment. The validation exercise orchestrates overflow compute capacity from public MEC infrastructure in response to simulated crowd-driven load spikes, and measures end-to-end latency, throughput, and handoff times for crowd management analytics workloads under peak conditions.

4.4.2 Task 4.2 - PPDR Scenario

This task realises and verifies the multi-level federation mechanisms in a simulated cross-border PPDR scenario, modelling first-responder communications that must maintain operational continuity while crossing administrative and national boundaries. The validation assesses end-to-end performance, policy enforcement under dynamic federated conditions, and the ability of the system to sustain mission-critical service guarantees throughout inter-domain mobility events.

4.5 WP5: Course, Final Thesis and Dissemination

WP5 encompasses tasks relating to mandatory activities of the doctoral program, as well as continuous dissemination activities, covering scientific publication of research contributions and open-source release of all developed components.

4.5.1 Task 5.1 - Curricular Classes

This task corresponds to the curricular courses of the PhD, which are Scientific Research Methodology, Thesis Project Preparation, and a Free Option, which was chosen to be Tutorial Practice. This task also encompasses the deliverables for each class.

4.5.2 Task 5.2 - Thesis Proposal

This task aims to define the objective of the thesis, resulting in a fundamented thesis proposal that includes a preliminary State of the Art (SotA) survey and a research statement (the present document).

4.5.3 Task 5.3 - Progress Reports

This task covers the annual progress reports stating the developments made over the respective year of research, identifying if the proposed milestones are being accomplished and possible adjustments to the research and development plans.

4.5.4 Task 5.4 - Final Thesis

This task covers the writing and public defence of the final PhD thesis document, synthesising contributions across all work packages into a coherent narrative.

4.5.5 Task 5.5 - Article Publishing

This task manages the submission and publication of scientific articles to top-tier telecommunications conferences and journals, covering the key architectural, implementation, and evaluation contributions produced across WPs2–4 (R5). The publication strategy is aligned with research milestone completion, aiming to publish at least one paper per WP.

4.5.6 Task 5.6 - Open-Source Releases

This task coordinates the release of the federation and orchestration components developed in WP3 as open-source contributions, enabling the broader research and operator community to deploy, validate, and extend the developed mechanisms.

4.6 Deliverables

Table 4.1: Research Deliverables

ID	Description	Due
TP	Thesis Proposal	Jun 2026
PR	Progress Report	Sep 2026
FT	Final Thesis document	Jun 2028
TD	Thesis Defence	Aug 2028

4.7 Milestones

Several milestones were identified to be completed throughout the project; these are shown in Table 4.2. In Figure 4.1, a Gantt chart is used to visualize the proposed workplan along the duration of the program.

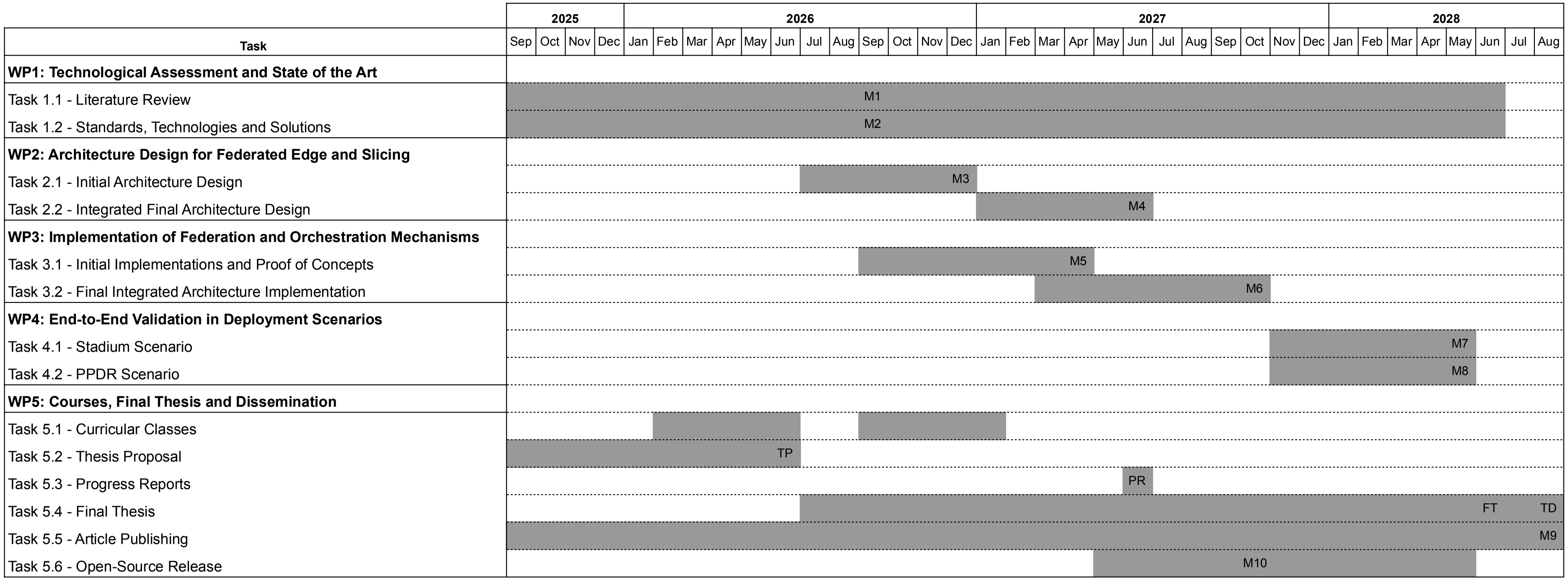

Figure 4.1: Workplan Gantt Chart

Table 4.2: Research Milestones

ID	Due	Description
M1	Sep 2026	SotA survey on MEC, network slicing, and multi-domain federation complete
M2	Sep 2026	Review of standardization and standard-compliant solutions complete
M3	Jul 2026	Initial architecture designs for slice mobility and inter-domain comms complete
M4	Oct 2026	Integrated final architecture design complete
M5	Oct 2027	Initial Proof of Concept (PoC) implementations complete and unit-tested
M6	Oct 2027	Final integrated architecture implemented and open-source released (R1, R2)
M7	May 2028	Smart Stadium scenario validation complete (R3)
M8	May 2028	PPDR cross-border scenario validation complete (R4)
M9	Aug 2028	$\geq$ 3 scientific publications in top-tier venues
M10	Aug 2028	Open-source framework release

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